

Sachit Ravikumar

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MSc Robotics graduate (University of Bristol) with expertise in **AI, Agents, machine learning, computer vision**. Skilled in **Python, C++, PyTorch, TensorFlow, Langgraph, SQL** for developing and deploying scalable AI systems. Passionate about building **end-to-end Software Systems** that solve real-world problems.

EDUCATION

MSc in Robotics, Distinction | University of Bristol | UK | 2023-2024

- **Key Modules:** Robotics Systems, Machine Vision, Intro to AI, Human-Robot Interaction, Bio-Inspired AI.

BEng in Computer Science and Engineering, First Class Distinction | Anna University | IN | 2019-2023

- **Key Modules:** Machine Learning, AI, Data Analytics, DBMS, Linear Algebra, Software Engineering.

PROJECTS

FixFlow - Autonomous AI Quoting Agent For Trade Businesses

- Built a production-grade autonomous AI agent for emergency trades businesses using **LangGraph** and **FastAPI**, handling the full quoting workflow - symptom diagnosis, job classification, dynamic pricing, negotiation, and authority enforcement, end-to-end in under 60 seconds with no human in the loop.
- Engineered a cyclic state machine with 11 nodes, deterministic safety guardrails, and integrations with **Civic MCP**(Calendar availability + prompt injection defence) and **Luffa**(customer interaction), with full node-by-node tracing via **LangSmith** and session persistence via **Supabase**.

Sign Language Detection

- Developed a **DeTr-based object detection pipeline** in **PyTorch** for real-time **sign language recognition**, with emphasis on **dataset curation, model generalisation, and benchmarking** (CNN, YOLO, Transformers) to enhance **accessibility and human-computer interaction**.
- Deployed the model on **AWS** using **Docker** and **FastAPI** for scalable, real-time inference and seamless cloud integration.

AI Nutrition Coach for Personalized Diet Planning

- Built a full-stack nutrition and meal planning web app using **React 18, Node.js, and PostgreSQL**, with an LLM agent (via OpenRouter) to generate personalised recipes, analyse URLs, and deliver weekly coaching insights based on user goals, allergies, and dietary preferences.
- Implemented secure **REST API** with auth, BMR/TDEE-based macro tracking, and photo-based meal logging with nutritional value estimation, deployed across **Vercel, Railway, and Supabase**.

TECHINCAL SKILLS

AI & Agents: LangGraph, LangSmith, PyTorch, TensorFlow, OpenRouter API's

Robotics & ML: ROS2, Gazebo, TensorFlow, PyTorch OpenCV, LiDAR integration, Docker

Languages & Frameworks: Python, C, C++, JavaScript, FastAPI, React, Node.js, Flask, C#, MATLAB

Cloud & Deployment: AWS ECS, Docker, Supabase, Vercel, Railway, Git

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Stealth Startup | April 2026 – Present | Remote

- Built and maintained a real-time voice AI pipeline using **Pipecat** with **Google Gemini Live API** for bidirectional streaming - integrating **ElevenLabs TTS** and **Google Cloud STT** for sub-second conversational latency across a production multi-tenant SaaS platform.
- Implemented **RAG** for AI agents using **LangChain** and **pgvector** (PostgreSQL); developed an edge voice agent on **Cloudflare Workers** with Durable Objects for stateful, low-latency **WebRTC** sessions at scale.
- Maintained **Docker + Kubernetes** deployment manifests and **CI/CD** pipelines for a multi-service monorepo; contributed to **React + TypeScript** dashboards for call analytics and real-time conversation monitoring with **Twilio, Stripe**, and CRM integrations.

Frontend Developer | TCR Innovation | January 2022 – March 2022 | Mumbai, India (Remote)

- Designed and deployed user-focused **React/JavaScript** platforms improving engagement by 30% - building responsive UIs, integrating **REST APIs**, and collaborating with stakeholders to translate requirements into production-ready web solutions.

AR Developer | XRBit | April 2021 – April 2022 | Chennai, India

- Built AR and 3D web experiences using **Three.js, Unity, and C#** - developing immersive cultural heritage applications, AR product visualisations, and interactive 3D scenes reaching 2,000+ visitors.